

Hardware plan for 25 riders at once

VR headsets · network · motion chairs · wind · water · safety · room

Bill of materials, diagrams, budget and operating plan

Prepared October 2026 · prices are estimates in USD (SAR at 3.75) — confirm with local suppliers

25 + 5 VR headsets (25 riders + 5 spares)	\$42k-79k Hardware budget (USD), excluding the chairs	4-5 shows/h ≈ 100-125 riders per hour with 5-8 min experiences	~175 m² hall + loading + control room
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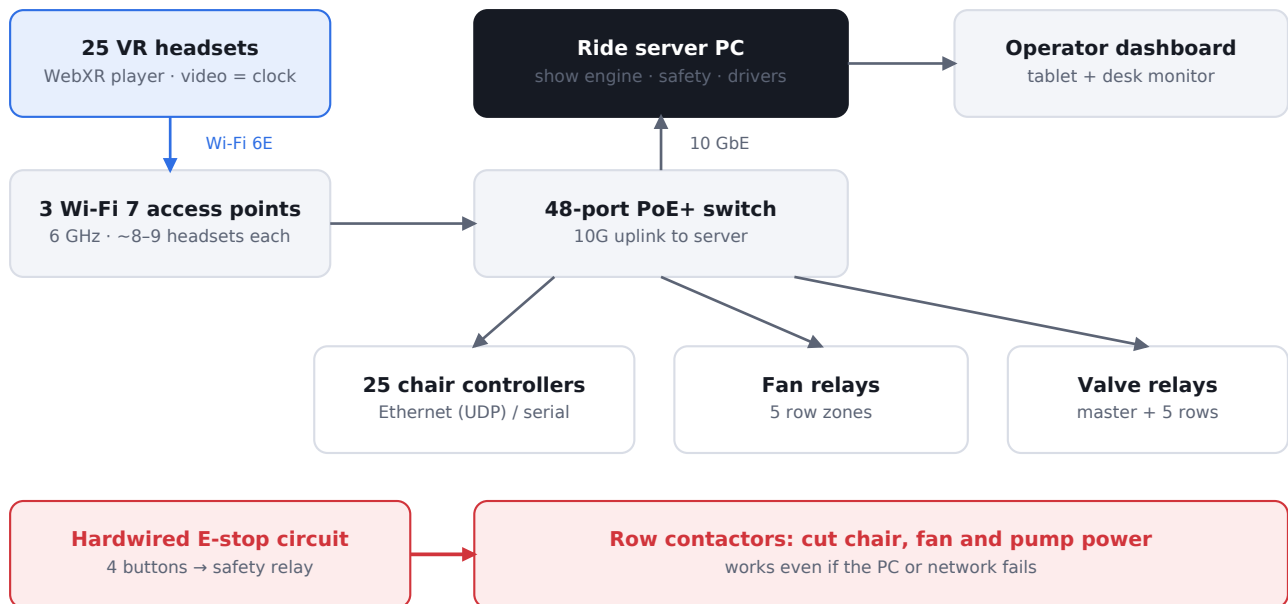
What this document covers

- Everything you need to buy so **25 people can ride at the same time**: headsets and rider kit, the Wi-Fi network and ride server, connecting and powering the motion chairs, the wind and water effects, the emergency-stop system, and the room.
- Quantities include sensible spares. Every line says **why** it is needed, so you can swap in local equivalents.
- Diagrams: system overview, hall floor plan, water system and emergency-stop circuit.
- The software (NASR VR Ride, already built) runs all 25 seats from one PC: each chair follows its own headset's video.

Your chairs. You already have motion chairs (xyz movement + vibration), so they are **not** in the budget. If you need more to reach 25, commercial single-seat VR motion chairs cost roughly \$3,000-\$8,000 each (≈ \$75,000-\$200,000 for 25).

1. How the system fits together

Every rider wears a standalone VR headset that plays the ride video. The headset tells the ride server where it is in the video ten times a second, and the server moves that rider's chair, fan and mist to match. The headsets need no cables. Everything else (chairs, fan and valve relays) is wired to one network switch. The emergency stop is a separate **hardwired** circuit that cuts power to the chairs and effects even if the PC or network fails.



Design assumptions

Item	Assumption used in this plan	If different
Group size	25 riders per show, all watching the same experience in sync	Software also supports fewer seats per show
Show length	5-8 minutes + ~5 minutes to load/unload and fit headsets	Longer shows → fewer riders/hour, more battery use
Video	360° stereo up to 5.7K, H.265 at ~30-40 Mbps	8K needs PC-VR headsets (much higher cost)
Chairs	Your existing chairs, 1 per rider, controller reachable by Ethernet or USB/serial	Send the chair model, and I'll adapt the driver
Power	3-phase supply available; chairs ~1-2 kW peak each (confirm on the nameplate)	Electrician sizes the board from real figures
Operating day	Up to 10-12 hours, staff of 1 operator + 3 attendants	

2. VR headsets & rider kit

Recommendation: Meta Quest 3 (512 GB). For watching 360° video, lens clarity matters more than anything else. Quest 3's pancake lenses are sharp edge to edge, it has a 3.5 mm headphone jack, Wi-Fi 6E for the uncrowded 6 GHz band, and since February 2026 every consumer Quest can be enrolled in Meta's device management (kiosk lock, remote setup) at no extra cost.

	Meta Quest 3 (512 GB) — recommended	Meta Quest 3S (128 GB) — budget
US price (since 19 Apr 2026)	\$599.99	\$349.99
Lenses / clarity	Pancake, sharp across the view, 2064×2208 per eye	Fresnel, softer edges, 1832×1920 per eye
Headphone jack	Yes, 3.5 mm	No (USB-C headphones or adapter needed)
Wi-Fi	Wi-Fi 6E (6 GHz)	Wi-Fi 6E (6 GHz)
30 units	\$18,000	\$10,500
Verdict	Best picture: riders notice sharpness in 360° video	Saves about \$7,500; acceptable for short rides

Why 30 headsets for 25 seats

- 5 spares (20%) cover drops, a cracked lens, a headset that won't connect, or one being cleaned. A show should never start with an empty seat.
- **Battery:** a Quest 3 runs about 2–2.5 h of video. With a hot-swap battery strap, staff swap a charged pack between shows in 10 seconds, so headsets stay in VR mode all day and charge fully overnight in the cabinet.
- **Hygiene:** every rider gets a disposable eye mask; facial interfaces are wiped and rotated (2 per headset); headsets + headphones go through the UV-C cabinet between groups.
- **Audio:** headphones, not the built-in speakers. With 25 headsets in one room, tiny timing differences between them sound like an echo.

Keep the water off the headsets. Quest headsets are not waterproof. Mist nozzles must be aimed at the hands, neck and lower face, never straight at the lenses, and the mist must be fine (see section 6).

3. Network & ride server

Twenty-five headsets streaming high-bitrate video at once is the most demanding part of the system. The plan keeps the headsets on their own clean 6 GHz Wi-Fi, spread over three access points, with a 10 Gb link from the server so the server is never the bottleneck.

Calculation	Value
Video bitrate per headset (5.7K 360° H.265)	30–40 Mbps
All 25 headsets playing	≈ 0.75–1.0 Gbps total
Per access point (3 APs, ~8–9 headsets each)	≈ 250–350 Mbps — comfortable for one 160 MHz 6 GHz channel
Server link	10 GbE (minimum 2.5 GbE)
Chair control traffic (25 chairs × 60 updates/s)	< 1 Mbps, wired, negligible

- Give the headsets their own **6 GHz-only SSID** with no internet and no guests. Put staff devices on 5 GHz, and chairs and relays on a separate wired network (VLAN).
- Mount the 3 access points on the ceiling above the side aisles (front, middle, back; see the floor plan) on **non-overlapping 160 MHz channels**. Turn transmit power down to medium, so each headset sticks to its nearest access point.
- Install the ride software on the backup PC with the same content and config. If the main PC fails, swap the network cable and carry on.
- Content tip: encode videos with faststart (see the software README). Shorter loading times mean shorter gaps between shows.

4. Motion chairs: control & power

The software sends each chair its target position (pitch, roll, yaw, heave, surge, sway, vibration) 60 times a second. It works with most chairs in one of three ways:

Your chair's controller...	Connect it like this	Extra hardware
Has an Ethernet port / runs its own motion software (FlyPT Mover, SimTools, vendor software)	Ride server → UDP over the wired network	None (Cat6 drop to each chair)
Is USB / serial only (Arduino, SMC3, Thanos AMC, many Chinese 9D chairs)	Serial-to-Ethernet converter at each chair, UDP mode	25 converters (optional line in the budget)
Uses a proprietary protocol	Send me the model + protocol document; I add a driver	Usually none

Power

- Check the nameplate of one chair for **peak** power. Typical 2-3-axis electric chairs draw 1-2 kW peak, which is roughly 25-50 kW for 25 chairs if all move hard at once. Because everyone watches the same video, **all chairs do move together**, so size for that.
- Use a dedicated 3-phase distribution board with a breaker per 2-3 chairs, 30 mA RCD protection, and the chairs split into 5 row circuits (rows 1-5). Each row passes through an E-stop contactor (section 7).
- Bolt every chair to the slab. Run cables in floor ducts or trenches, never loose across the floor. Use flexible, strain-relieved cable where it enters the moving part of the chair.
- Have a licensed electrician design and sign off the installation.

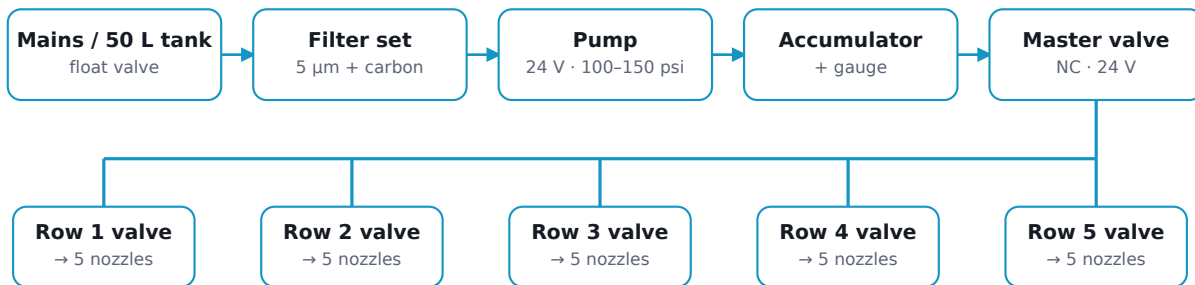
Before buying more chairs: make sure every chair has a seat belt or lap bar, a physical end stop on every axis, and overload protection. Software limits (already built in) are a second layer, not the first.

5. Wind effects

- **One compact fan per seat**, on a floor post 60-80 cm in front of the rider at chest height. A few big fans at the front would blast row 1 and leave row 5 with nothing.
- Wire the fans in **5 row circuits** switched by 2 × Shelly Pro 4PM relays (DIN rail, Ethernet). The software already speaks to Shelly, and the 3 spare channels can run extra effects later (scent, heat lamps).
- On/off fans are cheapest and work well. For variable wind (a breeze that builds to a gale), choose EC/DC fans with 0-10 V or PWM speed control; the software already outputs fan level 0-100%.
- The same post carries the mist nozzle (section 6), so you install one post per seat, not two.

6. Water effects (mist)

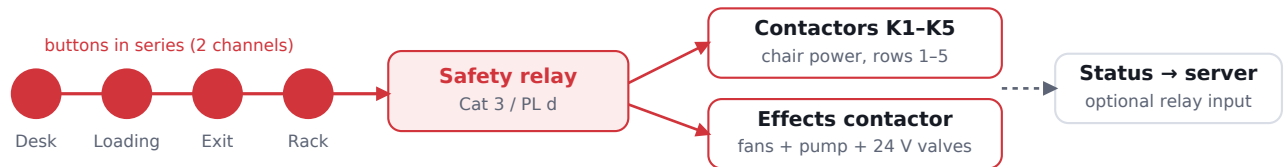
Use a **fine mist** from a 100–150 psi misting system. It gives a strong “splash” feeling, dries in seconds, and uses very little water: about 0.1 L per nozzle per minute of spraying. Normal garden sprinkler pressure would soak riders and headsets.



- **Normally-closed 24 V valves:** if power or the PC fails, the water stops. A master valve plus one valve per row means two valves must fail before water flows.
- **Anti-drip nozzles:** a check valve in each nozzle stops dripping after a burst.
- **Three layers of limits:** (1) the software caps each burst at 1.5 s, forces a gap between bursts and sets a maximum per show; (2) the Shelly relays have an auto-off timer; (3) the E-stop cuts valve power.
- **Hygiene:** use drinking-quality water and never let it stand warm. Flush the lines daily, clean the tank weekly, and add a UV steriliser if you use a tank.
- **Electrical:** only 24 V reaches the seats. Mains voltage stays in the control cabinet, and the pump sits in a drip tray.

7. Safety & emergency stop

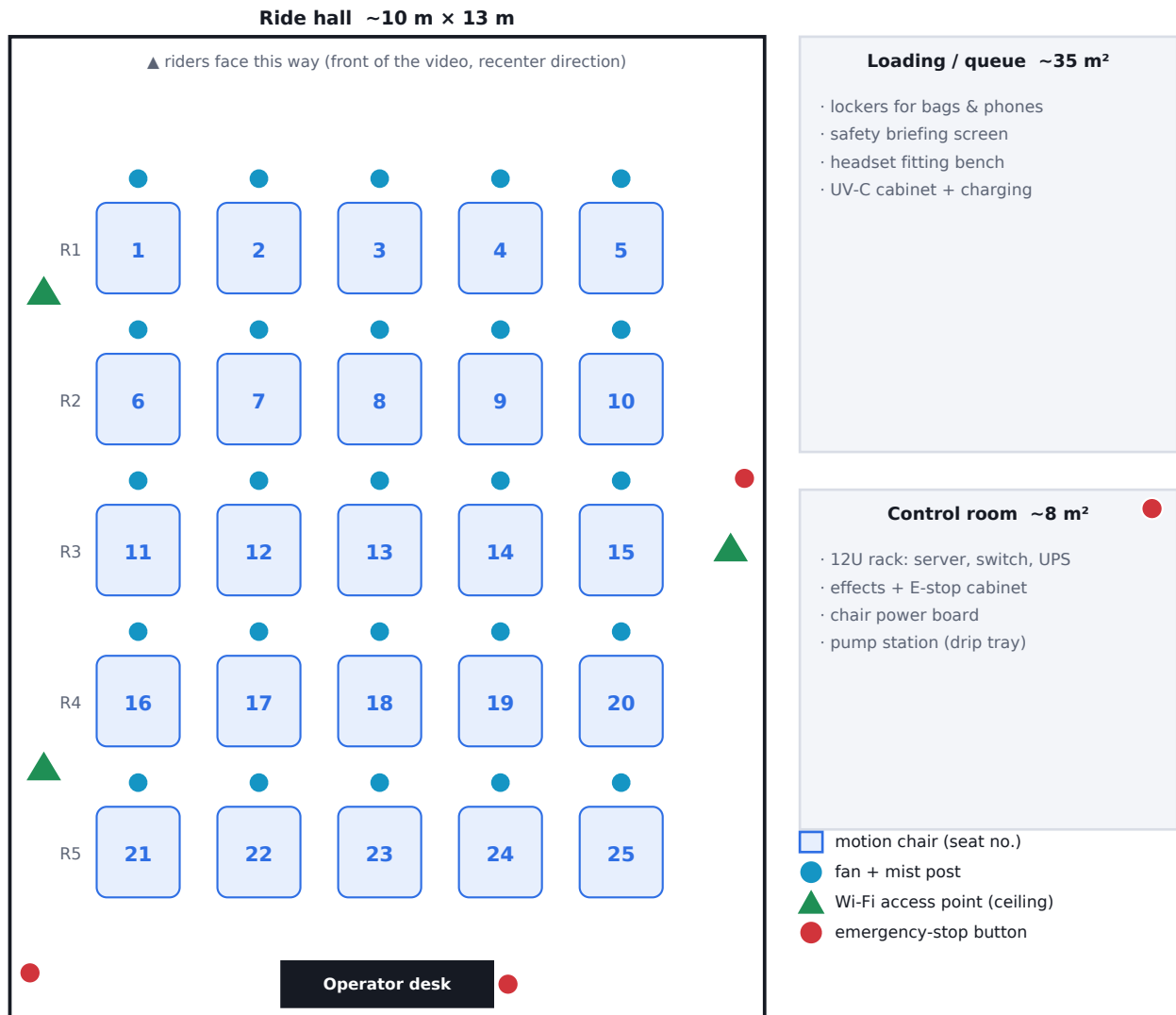
The software already parks every chair smoothly on E-stop, if a headset disconnects, or if a rider takes the headset off. It also cuts fans and water and shows a “Ride paused, please stay seated” message inside every headset. **A hardwired emergency-stop circuit is still mandatory**, because software can’t protect anyone if the PC itself has failed.



- 4 latching red mushroom buttons: operator desk, loading area, hall exit and the equipment rack. All are wired in series into a safety relay (Category 3 / PL d).
- The safety relay drops 5 row contactors (chair power) and 1 effects contactor (fans, pump, 24 V valves). Restarting needs a deliberate reset by the operator.
- Optional: wire a spare contact from the safety relay to a relay input, so the dashboard shows when the hardware E-stop has been pressed.
- 2 low-light cameras over the hall, shown on the operator desk. Riders can’t see the staff, so staff must be able to see every rider.
- Emergency lighting, clear exit routes, height/age/health rules at the entrance, a first aid kit and sick bags.

Get it approved. Ask the motion-chair maker for their safety requirements, and have the electrical design, E-stop circuit and water installation checked by a qualified engineer and the local civil-defence / entertainment authority before opening to the public.

8. Room & floor plan



Seat pitch ~1.6 m, row pitch ~2.0 m (check against your chair's footprint + motion clearance). Aisles ≥ 1.2 m on both sides.

Requirement	Recommendation
Floor area	Ride hall ~130 m ² (10 × 13 m) + loading/queue ~35 m ² + control room ~8 m ²
Floor	Concrete slab for chair anchoring; waterproof anti-slip vinyl on top; floor ducts for power and Cat6
Ceiling	≥ 2.7 m clear (access points, cameras, cable trays)
Cooling	25 riders + 25 chairs + equipment ≈ 8-12 kW of heat. Size the AC with an HVAC engineer and add dehumidification because of the mist
Lighting	Dim, even lighting (headset tracking needs some light); emergency lighting on separate circuits
Layout	Riders face the same wall: that's the "front" of the video. Flat floor; no screens or sight lines needed

9. Operating the hall

Step	What happens	Time
1. Queue	Safety briefing video; bags and phones into lockers; disposable eye masks handed out	during previous show
2. Seat	Riders sit, belts on; attendants fit headsets + headphones ($\approx$ 8 riders per attendant)	3-4 min
3. Check	Dashboard shows all 25 seats “ready · in VR · on head”; operator presses Start	30 s
4. Ride	Video, motion, wind and mist in sync; operator watches the cameras and dashboard	5-8 min
5. Exit	Chairs park automatically; headsets collected → wipe → UV-C; battery packs swapped as needed	2-3 min

Staff per shift: 1 operator (dashboard, E-stop, cameras) and 3 attendants (fitting, belts, cleaning).
Throughput is about 4-5 shows an hour, or 100-125 riders an hour.

Daily checklist

- Open: power on the rack, then the chairs. Run **Test chair** on every seat with nobody seated, and test the fans and mist with the dashboard test buttons.
- Check the battery levels on the dashboard; charged packs go to the swap tray.
- Close: headsets into the charging cabinet, flush the water lines, wipe all facial interfaces, check the logs for any seat faults.

10. Budget summary

Hardware estimates in USD, before VAT, shipping and import duty. Installation labour is included only where marked “installed”. The low figure excludes optional lines, the high figure includes them. Local prices in Saudi Arabia may be higher than US list prices.

Category	Low	High	Low (SAR)	High (SAR)
VR headsets & rider kit	\$24,350	\$34,260	91,312	128,475
Network, servers & control	\$6,840	\$14,150	25,650	53,062
Motion chairs: integration & power	\$3,500	\$11,250	13,125	42,188
Wind effects (fans)	\$2,354	\$5,420	8,828	20,325
Water effects (mist)	\$1,355	\$3,720	5,081	13,950
Safety & electrical	\$1,780	\$4,560	6,675	17,100
Room fit-out	\$1,950	\$5,200	7,312	19,500
Total hardware (excluding chairs)	\$42,129	\$78,560	157,984	294,600

Adjustments	Effect on total
Use Quest 3S instead of Quest 3	– about \$7,500 (plus ~\$10 per headset for USB-C headphones/adapters)
Buy 25 motion chairs (if not owned)	+ \$75,000 to \$200,000
Not included	HVAC, building works, decoration/theming, 360° content production or licences, permits, staff
Contingency	Add 10–15% for cables, fixings and things found on site

11. Full bill of materials

“Opt.” = only needed in some setups (see the reason). Unit prices are low-high estimates in USD.

VR headsets & rider kit

Item	Why / spec	Qty	Unit (USD)	Total (USD)
Meta Quest 3 (512 GB) headset	Pancake lenses, 2064×2208 px per eye, Wi-Fi 6E, 3.5 mm audio jack. 25 in use + 5 spares.	30	600–700	\$18,000–\$21,000
Battery comfort strap with hot-swap battery	e.g. BoboVR M3 Pro class. Balances weight on the head and roughly doubles run time.	30	70–110	\$2,100–\$3,300
Extra swappable battery packs	Charged packs are swapped between shows, so headsets never leave the hall to charge.	30	35–55	\$1,050–\$1,650
Multi-bay charging cabinet	Lockable, charges 30 headsets + battery packs overnight; USB-C PD.	1	600–1,500	\$600–\$1,500
Wipeable facial interfaces (PU/silicone)	2 per headset, so one set is always cleaned and dry.	60	15–30	\$900–\$1,800
Disposable hygiene eye masks	One per rider. Starter stock of about 3 weeks.	3,000	0.05–0.12	\$150–\$360
UV-C headset sanitising cabinet	Disinfects headsets + headphones between groups (60–90 s cycle).	1	800–3,000	\$800–\$3,000
On-ear wired headphones, short cable	Wipeable leatherette cushions. Private audio avoids echo between 25 headsets.	30	20–45	\$600–\$1,350
Spare face/ear cushions, lens cloths, cleaning wipes	Alcohol-free lens wipes, microfibre cloths.	1	150–300	\$150–\$300

Network, servers & control

Item	Why / spec	Qty	Unit (USD)	Total (USD)
Ride server PC	8-core CPU, 32 GB RAM, 2 TB NVMe, 10 GbE network card. Runs the NASR VR Ride software.	1	1,500–2,500	\$1,500–\$2,500
Backup PC (cold standby)	Mini PC with the same software and content; swap in within minutes.	1	500–900	\$500–\$900
Wi-Fi 7 access points with 6 GHz	UniFi U7 Pro or equivalent, ceiling-mounted, PoE. 3 in use + 1 spare.	4	190–260	\$760–\$1,040
Gateway / router	Separate networks for headsets, chairs and staff.	1	200–400	\$200–\$400
Managed 48-port PoE+ switch, 10G uplink	Powers the access points; Ethernet to all 25 chairs and effect controllers.	1	900–1,500	\$900–\$1,500
10G DAC cable / SFP+ module	Server ↔ switch.	1	30–60	\$30–\$60
12U rack, patch panel, PDU	Lockable, in a ventilated control room.	1	400–800	\$400–\$800
UPS 2200 VA, line-interactive	Keeps server, switch and access points up through power dips.	1	500–1,000	\$500–\$1,000
Operator tablet	Runs the dashboard: load / start / stop, live seat status.	1	350–600	\$350–\$600
Operator monitor + keyboard	Fixed dashboard at the operator desk.	1	200–350	\$200–\$350
Cat6 cabling, ~35 drops, installed	Floor/trunk to each chair, access points, effects cabinet.	1	1,500–3,500	\$1,500–\$3,500
Serial-to-Ethernet converters (opt.)	Only if your chair controllers are USB/serial only (e.g. USR-TCP232 class, UDP mode).	25	25–60	\$625–\$1,500

Motion chairs: integration & power

Item	Why / spec	Qty	Unit (USD)	Total (USD)
Chair power distribution board	3-phase board, a breaker per 2-3 chairs, 30 mA RCDs, installed by a licensed electrician.	1	3,000-8,000	\$3,000-\$8,000
Floor anchoring kits	Bolts the chair base to the slab so it can't walk.	25	20-50	\$500-\$1,250
Seat belts / lap bars (opt.)	Only if your chairs don't already have them.	25	30-80	\$750-\$2,000

Wind effects (fans)

Item	Why / spec	Qty	Unit (USD)	Total (USD)
Compact high-velocity fans, 150-200 mm	One per seat, aimed at face/chest from 60-80 cm. 25 + 3 spares.	28	30-70	\$840-\$1,960
Floor post per seat (fan + mist nozzle)	Steel post with bracket, bolted to the floor in front of each chair.	25	40-100	\$1,000-\$2,500
Shelly Pro 4PM (4-channel relay, DIN)	Switches the 5 fan rows; 3 spare channels.	2	107-130	\$214-\$260
Fan wiring and cable ducts	One circuit per row.	1	300-700	\$300-\$700

Water effects (mist)

Item	Why / spec	Qty	Unit (USD)	Total (USD)
Misting pump, 24 V diaphragm, 100-150 psi	Fine mist, not a soak. 1 + 1 spare.	2	80-250	\$160-\$500
50 L food-grade tank with float valve	Or direct mains feed with pressure regulator + backflow preventer.	1	80-200	\$80-\$200
Filter set: 5 µm sediment + carbon, gauge, regulator	Clean water = no clogged nozzles.	1	80-200	\$80-\$200
Accumulator / small pressure tank	Instant mist on demand, fewer pump starts.	1	50-120	\$50-\$120
Normally-closed solenoid valves, 24 V DC	1 master + 5 rows + 1 spare. Close automatically on power loss.	7	25-60	\$175-\$420
Misting nozzles with anti-drip check valve	0.2-0.3 mm. One per seat + spares.	30	3-8	\$90-\$240
PE tubing 3/8" + 1/4", push-fit fittings	Main line along each row, short drop to each post.	1	200-500	\$200-\$500
Shelly Pro 3 (dry-contact relays, DIN)	Switches 24 V to the valves; auto-off timer as a hardware backstop.	2	110-150	\$220-\$300
24 V DC DIN-rail power supply, 10 A	Valves and pump. Keeps mains voltage away from the wet area.	2	50-120	\$100-\$240
Drip trays / drain covers	Under the pump station and manifold.	1	200-600	\$200-\$600
UV water steriliser (opt.)	Recommended if a tank is used (stops bacteria growing in standing water).	1	150-400	\$150-\$400

Safety & electrical

Item	Why / spec	Qty	Unit (USD)	Total (USD)
Safety relay (Cat 3 / PL d)	e.g. Pilz PNOZ or Schneider Preventa. Monitors the E-stop circuit.	1	200-450	\$200-\$450
Emergency-stop buttons (red mushroom, latching)	Operator desk, loading area, hall exit, equipment rack.	4	40-90	\$160-\$360
Contactors, one per chair row	Cut chair power on E-stop. Hardwired, independent of software.	5	100-250	\$500-\$1,250
Effects + E-stop control cabinet	DIN rail, terminals, breakers, labelling.	1	500-1,200	\$500-\$1,200
Low-light IP cameras over the hall	Staff can watch every rider from the operator desk.	2	60-200	\$120-\$400
Emergency lighting + exit signs	Hall is kept dim; riders must be able to get out in a power cut.	1	200-600	\$200-\$600
First aid kit, sick bags, rules signage	Motion sickness happens. Plan for it.	1	100-300	\$100-\$300

Room fit-out

Item	Why / spec	Qty	Unit (USD)	Total (USD)
Waterproof anti-slip vinyl flooring, ~130 m ²	Mist + moving chairs; must be easy to dry and clean.	130	15-40	\$1,950-\$5,200

12. Next steps

#	Action	Who
1	Send the chair model, controller type and nameplate power. I'll confirm the connection method and add a driver if needed	You → me
2	Order 2 headsets + 1 access point first; test your real videos on them at the venue	You
3	Electrician + HVAC engineer survey the hall (power, cooling, floor ducts)	You
4	Install 1 full row (5 seats) with fans + mist; run the software end to end for a week	You + me
5	Roll out the remaining 4 rows; safety inspection; staff training; soft opening	You

Sources for prices

Meta Quest pricing update, 19 April 2026 (Quest 3 512 GB \$599.99, Quest 3S 128 GB \$349.99): meta.com/blog/update-meta-quest-pricing · Meta for Work update, Feb 2026 (all consumer Quest 3/3S enrollable in Horizon Managed Solutions at no cost): forwork.meta.com · Shelly Pro 4PM: us.shelly.com · Ubiquiti U7 Pro launch price \$189. Other prices are typical 2026 market ranges; confirm with your local suppliers.